

TITLE PAGE

- Problem Statement ID:

26143
- Problem Statement Title:

Leveraging satellite imagery to determine Oil spills at sea along with AIS data correlations to identify vessel responsible for the spill.
- Theme:

Disaster Management
- PS Category:

Software
- Project Name:

MARIS-TRACER (Multi-Modal Oil Spill Source Attribution)
- Organization / Ministry:

National Technical Research Organisation (NTRO)
- Team ID:

_____ (Left blank as per SIH guidelines)
- Team Name:

Team MARIS

MARIS-TRACER

Decision Support Framework

1. Deep SAR Vision

U-Net capillary wave dampening segmentation on Sentinel-1 SAR.

2. 4D Hydrodynamics

Backward particle tracking (OpenDrift) under wind & currents.

3. AIS Correlation

Spatiotemporal vessel trajectory query & candidate ranking.

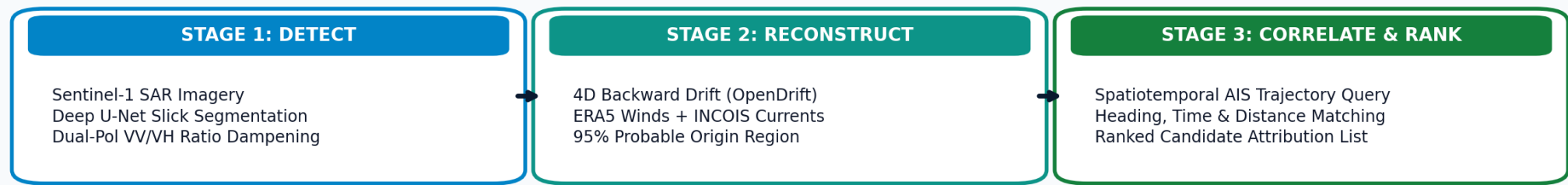
4. Decision Support

Evidence-backed report generation for Coast Guard / NTRO.

PROPOSED SOLUTION - FROM DETECTION TO SOURCE ATTRIBUTION

Idea Title: MARIS-TRACER

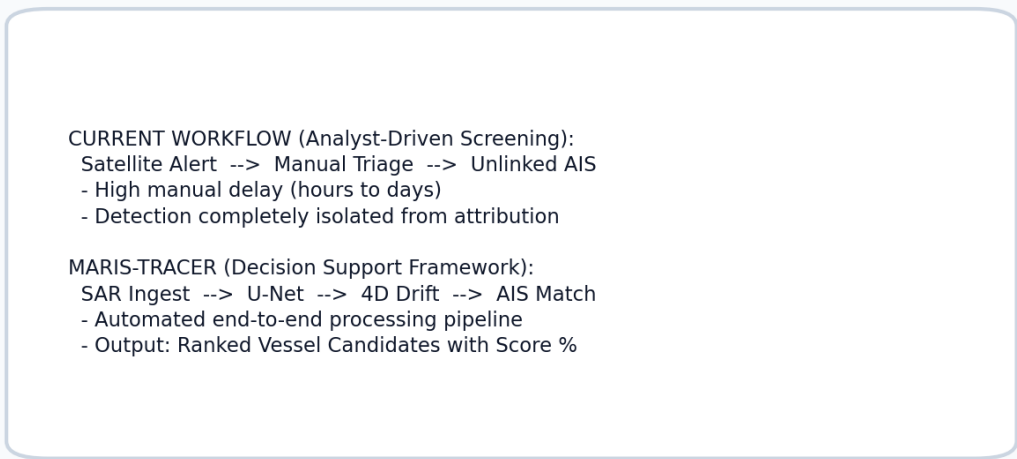
PROPOSED 3-STAGE ATTRIBUTION PIPELINE



FALSE-POSITIVE LOOK-ALIKE REJECTION GATING



WORKFLOW COMPARISON: CURRENT VS MARIS-TRACER



TECHNICAL APPROACH - SYSTEM ARCHITECTURE & SCOPE

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SYSTEM SCOPE: HACKATHON MVP VS ADVANCED PHASE

HACKATHON MVP (36-Hour Core - MUST WORK)

- Sentinel-1 GRD SAR Ingestion + SNAP Speckle Filter
- PyTorch U-Net Deep Learning Slick Segmentation
- 2D/3D Backward Particle Drift (ERA5 + INCOIS Currents)
- Spatiotemporal AIS Trajectory Bounding Query (PostGIS)
- Explainable Proximity Ranking + Interactive GIS Dashboard

ADVANCED PHASE (Post-Hackathon / If Time Permits)

- Bi-Temporal OSCD Change Detection & Multi-sensor Fusion
- 10,000 Particle Monte Carlo Ensemble Dispersion
- Dark-Ship Kinematic Dead Reckoning & Wake Detection
- Automated PDF Forensic Evidence Report Generation
- Real-Time Live NMEA/AIS Stream Parser Integration

TEAM EXPERTISE & MODULE OWNERSHIP MATRIX

Role / Expertise	Primary Responsibility	Tech Stack & Core Deliverable	Hackathon Owner
ML / CV Lead	SAR Preprocessing & U-Net Slick Segmentation	PyTorch, OpenCV, GDAL, Albumentations	Member 1
Hydrodynamics Lead	Ocean Current Forcing & Particle Backtracking	OpenDrift, PyGNOME, NetCDF4, SciPy	Member 2
Maritime GIS Lead	AIS Trajectory Indexing & Spatial Queries	PostgreSQL/PostGIS, GeoPandas, H3	Member 3
Full-Stack Lead	Backend API, Interactive Map UI & Report Engine	FastAPI, React, Mapbox GL JS, Leaflet	Member 4 & 5

FEASIBILITY & VIABILITY - DATA STRATEGY & RISK MITIGATION

Idea Title: MARIS-TRACER

DATA VIABILITY: DEMONSTRATION-FEASIBLE

- REAL / PUBLIC OPEN-ACCESS FEEDS:
- Sentinel-1 SAR GRD (ESA Copernicus API)
 - ERA5 Hourly 10m Surface Winds (ECMWF CDS)
 - INCOIS OSFS Daily Surface Current Velocity Grids
- DEMONSTRATION FALLBACK STRATEGY:
- Public/Historical AIS Datasets (NOAA MarineCadastre / GFW)
 - Synthetic AIS Vessel Trajectory Generator (for live testing)
 - Ensures 100% reliable 36-hr hackathon prototype demo.

DEPLOYMENT & HARDWARE REQUIREMENTS

- ZERO VESSEL HARDWARE REQUIRED:
- Operates entirely via remote satellite sensing and AIS.
 - No physical sensors or equipment installed on vessels.
- COMPUTE & HOSTING STACK:
- Single workstation / cloud instance with NVIDIA GPU.
 - 1-Click Docker Compose deployment containerizing:
FastAPI Backend + PostgreSQL/PostGIS + React WebGIS UI.

KEY TECHNICAL RISKS & PRACTICAL MITIGATION STRATEGIES

RISK 1: Look-Alike Slicks

Challenge:
Algae, biogenic films & calm waters dampen radar waves like oil slicks.

Mitigation:
Dual-Pol VV/VH ratio + GLCM texture + wind thresholding (> 2 m/s).

RISK 2: Dark Ships (No AIS)

Challenge:
Perpetrators disable AIS transponders (~6% of global fleet).

Mitigation:
Kinematic dead reckoning from past waypoints + SAR target detection.

RISK 3: Hydrodynamic Drift Error

Challenge:
Ocean current turbulence & wind gust forecasting uncertainties.

Mitigation:
Monte Carlo ensemble deriving probabilistic 95% origin region.

IMPACT & BENEFITS - EVALUATION PLAN & STRATEGIC VALUE

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END-TO-END 6-STEP HACKATHON DEMO FLOW



EVALUATION PLAN & TARGET METRICS

Module	Target Metric	Target Goal
Slick Segmentation	Dice / mIoU	>= 0.78
Look-Alike Rejection	False Positive Rate	<= 12%
Origin Drift Error	Median Error Distance	<= 4.5 km
Vessel Attribution	Top-3 Recall	>= 85%

MULTI-DIMENSIONAL STRATEGIC BENEFITS

ENVIRONMENTAL IMPACT

Accelerates spill containment boom deployment, protecting sensitive coastal ecosystems & mangroves.

OPERATIONAL EFFICIENCY

Replaces hours of manual screening with automated candidate ranking in under 3 minutes.

ENFORCEMENT PRIORITIZATION

Provides decision support for Coast Guard / NTRO to prioritize vessel inspections.

RESEARCH & REFERENCES - SCIENTIFIC & GOVERNANCE FOUNDATIONS

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SCIENTIFIC & GOVERNANCE REFERENCE FOUNDATIONS

1. SAR AI & DEEP LEARNING PAPERS

- Lai et al. (2024). Deep Learning for SAR Oil Spill Segmentation & OSCD. IEEE TGRS.
- Chen et al. (2023). DGNet: Distribution-Guided SAR Feature Extraction. IEEE TGRS.
- Moon et al. (2024). DAKTer Diffusion Augmentation for SAR Segmentation. arXiv:2412.08116.

2. HYDRODYNAMICS & LAGRANGIAN DRIFT

- Dagestad et al. (2018). OpenDrift: Open-source Trajectory Framework. Geosci. Model Dev.
- NOAA OR&R (2023). PyGNOME: Operational Oil Spill Modeling Environment.
- Stokes Drift & Wind Leeway Formulations for Surface Pollutant Backtracking.

3. MARITIME LAW & GOVERNANCE

- IMO (2024). MARPOL Annex I: Regulations for Prevention of Pollution by Oil.
- UNCLOS Part XII (1982). Protection and Preservation of Marine Environment. UN.
- EMSA (2023). CleanSeaNet Satellite Monitoring Service in European Waters.

4. NATIONAL DATA & STRATEGIC ASSETS

- INCOIS (MoES, Govt. of India). Ocean State Forecast System (OSFS) High-Res Currents.
- ESA Copernicus Hub. Sentinel-1 Synthetic Aperture Radar (SAR) Level-1 GRD.
- NOAA MarineCadastre & Global Fishing Watch. Historic AIS Vessel Tracking Datasets.